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Development of a centralized progress management system using lean thinking and efforts to improve operations

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Abstract

In this study, we improved MRI room operations by applying Lean Thinking principles with a strong emphasis on leadership and ICT integration. The primary objectives were to enhance MRI appointment slot utilization, reduce patient waiting times, and minimize referrals to external facilities. Lean methodologies such as Kaizen, 5 S, Value Stream Mapping, Standard Work, Spaghetti Chart, and Root Cause Analysis were used to identify inefficiencies and implement structured improvements. A centralized ICT-based management system was introduced to enable real-time tracking and enhance workflow coordination. Leadership played a key role in fostering staff engagement and continuous improvement. MRI workflows were analyzed using Lean tools to identify operational bottlenecks and streamline processes. A structured improvement approach was developed through brainstorming sessions and weekly short meetings to refine management practices. The ICT system facilitated real-time monitoring of patient progress and scheduling adjustments. The 5 S methodology was employed to improve workspace organization, while Kaizen supported incremental daily improvements based on staff input. Following these interventions, MRI appointment reservations increased by 26% and completed examinations grew by 19%, with further daily management refinements contributing additional improvements of 34% and 28%, respectively. Patient waiting times were significantly reduced from an average of 45 days to 5 days, and monthly external referrals for MRI examinations declined from 300 to 50. These outcomes demonstrate that the integrated application of Lean Thinking and ICT not only substantially enhanced operational efficiency, reduced patient delays, and optimized resource utilization but also improved patient access and workflow coordination. Leadership, combined with a supportive and collaborative culture, played a key role in empowering team members to engage in continuous improvement. Open communication, the active valuation of staff input, and the delegation of responsibilities contributed to establishing a sustainable model for process enhancement. Although challenges such as the need for continuous adjustments in daily management and the complexity of coordinating staff schedules while maintaining patient care remained, these results confirm that Lean Thinking originally developed in manufacturing can be effectively adapted to healthcare settings to enhance productivity, service accessibility, and overall operational performance.

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Keywords Lean thinking, MRI, Quality improvement, Kaizen, Radiology, Business process reengineering (BPR), Total quality management (TQM)

Introduction

Lean Thinking is a method of operational improvement based on the Toyota Production System. In 2016, Klein et al. investigated the utilization of Lean Thinking to improve the throughput of computed tomography (CT) scans in emergency medical care, leading to its application in the healthcare field [1, 2]. Klein et al. conducted a behavioral analysis and formulated an improvement plan based on Lean Thinking for the operating system, focusing on the workflow from ordering a non-contrast-enhanced CT scan in emergency medical care to the completion of the scan and the patient's return to the examination room. This approach successfully enhanced throughput efficiency [1].

Despite the demonstrated effectiveness of Lean Thinking in emergency medical care, challenges remain in improving daily operational efficiency in medical imaging departments, such as MRI examination departments. These departments frequently encounter prolonged waiting times, limited resource availability, and a reliance on outsourcing urgent examinations to external facilities. Such inefficiencies negatively impact patient experience, delay treatment, and impose financial burdens on healthcare institutions. Addressing these operational inefficiencies is essential to meet increasing patient demand, minimize unnecessary expenditures, and enhance the overall quality of care. However, existing research on Lean Thinking has primarily focused on emergency settings, leaving its application to routine operations in MRI departments largely unexplored.

This study explores the application of Lean Thinking beyond emergency care settings to improve both throughput and operational efficiency in daily medical imaging workflows. The MRI examination department of the Tokyo Women's Medical University Hospital, which faced operational inefficiencies, was selected as the target for improvement.

At this facility, MRI examinations were conducted using six units—four in the outpatient center and two in the ward examination room. However, limitations in scheduling flexibility made it difficult to conduct examinations at necessary times. In urgent cases, outsourcing to external facilities was often required, leading to an increased burden on institutional management due to the rising number of outsourced examinations.

To address these challenges, this study applies Lean Thinking to routine MRI operations, an area where operational inefficiencies are both prevalent and impactful. By incorporating Lean tools and leveraging Information and Communication Technology (ICT) to manage workflow

and resource allocation, this study aims to develop and validate a sustainable model for improving daily MRI operations and throughput. The integration of ICT is intended not only to streamline processes but also to provide real-time data for enhanced decision-making. Through this approach, the study contributes to both the academic discussion on Lean Thinking in healthcare and the development of practical solutions to critical challenges in medical imaging departments.

A review of existing studies on facility operation systems and throughput improvement using Lean Thinking in MRI departments revealed a lack of research in this area [3]. Given this gap, the study investigates whether Lean Thinking can be effectively utilized to enhance operations and throughput in daily MRI examination workflows. Furthermore, recognizing the potential of ICT in optimizing these processes, a new ICT system was developed to facilitate operational improvements and efficiency gains.

Based on this background, this study conducted a behavioral analysis and developed an improvement plan using Lean Thinking in the MRI department's daily operations. The goal was to enhance the utilization rate of MRI examination slots, reduce the number of waiting days for reservations, and minimize the outsourcing of examinations to external facilities.

In 2015, the Ministry of Health, Labour and Welfare issued a proposal titled "Healthcare 2035." The proposal's content is not a generalized list of existing initiatives but a vision 20 years into the future. To achieve this objective, the basic principles and values that should be respected in healthcare and the direction of change required are discussed from the perspective of 2035 [4, 5].

Within this "Healthcare 2035 Proposal," "Lean Healthcare: Enhancing the Value of Health Care" is listed as the core pillar [4]. This "lean healthcare" is a way of thinking that applies the concept of lean thinking to healthcare. Therefore, providing high-value services at lower costs is necessary, and this is positioned as lean healthcare" [4].

The Ministry of Health, Labor, and Welfare aims to build a world-class sustainable healthcare system by 2035. Based on the idea of "better medical care at a lower price," the concept of "lean healthcare" is expected to become an important initiative in Japan's healthcare in the future.

Dedicated literature review

Dr. H.L. Bhaskar has contributed extensively to the academic and practical development of Business Process Reengineering (BPR), Lean Thinking, and Total Quality

Management (TQM), particularly within the context of Industry 4.0 and manufacturing. Across nine publications, he builds a comprehensive body of knowledge integrating radical process transformation with continuous improvement philosophies and modern digital tools.

His most recent work proposes the B2Lean framework, a hybrid model that fuses the structural overhaul of BPR with the operational refinement of Lean Thinking, showing measurable benefits in cycle time, defect reduction, and process flexibility [6]. In parallel, Bhaskar and his co-authors explore how process mining tools crucial in Industry 4.0 can enhance real-time visibility and process diagnostics. This techno-functional perspective ensures that both technical capacity and business usability are considered when selecting tools [7].

In a related study, Bhaskar evaluates a wide range of BPM technologies (including RPA, process modeling, and optimization engines) in light of their fit for Industry 4.0 settings. He identifies successful adoption factors such as interoperability, scalability, and analytic capabilities [8].

Bhaskar's foundational works address the conceptual framing of BPR. He argues for BPR not as a one-off restructuring tool, but as a sustained, process-based management practice supported by IT, leadership, and change readiness [9, 10]. While the integration of ICT can significantly enhance process efficiency and data visibility in healthcare settings, it is important to acknowledge its inherent limitations. Bhaskar critically argues that ICT alone cannot drive meaningful or sustained transformation without concurrent changes in organizational structure, workflows, and culture [10]. Over-reliance on technology may lead to automation of inefficient processes, resulting in limited performance gains. Moreover, ICT systems often fail to capture the tacit knowledge and human interactions that are crucial to high-quality service delivery. Thus, for ICT implementation to be truly effective, it must be embedded within a broader framework of process reengineering and organizational alignment [10]. This perspective underscores the need to view ICT not as a standalone solution, but as an enabler that must be integrated with human-centered and systemic change efforts.

He also emphasizes the strategic synergy between BPR and quality systems, particularly TQM, showing how radical process reengineering can complement and amplify quality-driven continuous improvement [11–13]. Building on evidence that Total Quality Management (TQM) significantly strengthens a firm's market orientation—and that both, in turn, are positively associated with organizational performance—Bhaskar suggests a mechanism that directly supports the sustainability of Lean transformations [11]. TQM's emphasis on standardized problem solving, continuous improvement, and leadership

supported quality routines institutionalizes behaviors that keep Lean practices from reverting to pre change habits. Simultaneously, market orientation keeps the definition of “value” continuously calibrated to evolving patient/customer needs and competitive signals, ensuring that Lean's waste reduction and flow improvements remain strategically relevant rather than purely internal efficiency plays. Together, TQM builds the durable cultural and procedural backbone for daily Lean work, while market orientation supplies the outward facing feedback loop that preserves alignment and renews improvement targets an interplay consistent with the empirically observed links among TQM, market orientation, and performance.

Additionally, Bhaskar empirically validates the interconnection between TQM, market orientation, and firm performance. His findings support the notion that market-informed TQM initiatives drive performance more effectively [11]. He also constructs a conceptual maturity model for TQM that serves as a platform for later reengineering or digitalization [14].

Collectively, these works form an integrative view of BPR—not as an isolated practice but as a dynamic, scalable methodology aligned with organizational performance, digital tools, and quality systems.

In addition, recent research cited in this study has demonstrated the effectiveness of Lean Thinking in specific diagnostic contexts. For example, Klein et al. (2016) applied Lean strategies to emergency CT workflows, while Özçelik et al. (2016) studied Lean applications in MRI-Tomography centers [1, 2]. Furthermore, Womack's foundational work on Lean Production [15] and later applications in radiology scheduling and catheterization laboratories [16, 17] highlight the method's cross-sector adaptability. These studies underscore that Lean Thinking can enhance operational efficiency, reduce waste, and improve service quality not only in manufacturing but also in healthcare.

Importantly, leadership and organizational culture have been identified as key enablers of sustainable Lean implementation. Liker emphasized the development of leadership at all levels [18], and Tortorella et al. showed that task-oriented and relational leadership behaviors are critical during various Lean phases [19]. These findings are directly reflected in this study's implementation, which incorporated Daily Management, cross-functional team structures, and ICT solutions to create a replicable and scalable improvement model in MRI operations.

Materials and methods

Conventional operations improvement methods and lean thinking

Previous studies on Lean Thinking in healthcare, such as those by Klein et al. and Özçelik et al., have

predominantly focused on applying Lean Thinking within specific diagnostic contexts, such as emergency CT scans, and often combined Lean with Six Sigma approaches [1, 2]. In contrast, this study explores its application to daily MRI operations in non-urgent settings, demonstrating its potential as a sustainable method for continuous workflow improvements and waste elimination. Originally proposed by J. P. Womack in 1990, Lean Thinking classifies all work processes based on whether they add value. The primary objective is to systematically eliminate waste and enhance productivity. The aim is to eliminate waste thoroughly and improve productivity [15]. In the field of healthcare, this concept can be used to improve patient waiting time, treatment delays, operational efficiency, cost, processing capacity, quality of medical care, and work environment.

While Lean Thinking is widely employed in both the automotive and healthcare industries, the specific tools and methodologies utilized differ depending on the sector. A comparative analysis of representative tools and their respective applications is provided in Table 1.

In this study, four analytical tools were employed: Value Stream Mapping, Spaghetti Chart, Standardized Work, and Root Cause Analysis. Following this analysis, an improvement plan was developed through brainstorming sessions. To sustain continuous Kaizen, daily management practices were implemented to support ongoing enhancements. Additionally, the 5 S methodology was applied to execute and refine the improvement plan.

Traditional operational improvement methods primarily focus on reducing operation time, which constitutes only a small fraction of total work time. Given that value-adding tasks typically account for less than 5% of total work time, even a 50% reduction in this category results in only a marginal decrease in overall work duration [16] (Fig. 1). Lean Thinking, in contrast, prioritizes the reduction of non-value-adding activities, leading to a substantial decrease in total operation time. This approach facilitates the delivery of high-quality services while significantly improving performance [17]. Furthermore, this study distinguishes itself by incorporating a centralized ICT system to enable real-time MRI schedule monitoring and optimize resource utilization. This structured framework presents a replicable model for other healthcare institutions seeking to enhance operational efficiency.

Lean thinking cycle diagram and action plan

The Lean Thinking framework consists of five key steps: (1) determining evaluation indices, (2) visualizing the workflow, (3) creating a flow diagram and improvement plan, (4) implementing the improvement plan, and (5) evaluating and refining the Lean Thinking cycle. The corresponding action plan is outlined in Table 2.

In Lean Thinking, continuous evaluation and iterative improvements are essential. Implementing an improvement plan is not a one-time action; rather, it involves an ongoing process of assessing outcomes, identifying new areas for enhancement, and achieving higher operational efficiency Fig. 2.

Operation improvement method based on lean thinking

Determination of evaluation indexes

Establishing evaluation indices is crucial for setting the direction of operational improvement and objectively measuring progress. In the initial phase of formulating an operational improvement plan, the following indices were identified to assess current inefficiencies and prioritize improvement areas:

- Actual number of examinations, reservation slots, and operating rate.
- Number of waiting days for examination.
- Number of examination orders for other facilities.

Lean Thinking fosters a continuous improvement cycle. As the process advances, additional evaluation indices are introduced to further analyze and address specific inefficiencies. The details of this iterative approach are discussed in subsequent sections.

Visualization of workflow

Following the determination of evaluation indices, workflow visualization provides an objective assessment of existing inefficiencies. Four tools were employed in this study to achieve workflow visualization:

- Value Stream Mapping: This diagrammatic approach details all hospital processes, including reception, examination, and billing, capturing essential metrics such as cycle time, patient flow, and information flow. By utilizing Value Stream Mapping, as illustrated in Fig. 3, the current workflow was analyzed, and inefficiencies were identified [16, 17].
- Standardized Work: This methodology structures work processes into distinct elements to facilitate analysis. Factors such as procedural steps, staff roles, locations, technology, safety, patient guidance, and cycle time are considered. By incorporating patient and staff schedules into Standardized Work, as depicted in Fig. 4, delays, workflow inefficiencies, and sources of wasted time were identified.
- Spaghetti Chart: This visualization tool maps staff movement patterns within examination rooms and operational areas. By plotting these movements on a facility diagram, unnecessary or inefficient pathways were highlighted, enabling targeted improvements to

Table 1 Lean tools and methods used in automotive and healthcare yellow (no1-8): used in this study

	Tool/Method	Automotive	Healthcare
1	Value Stream Mapping (VSM)	Visualize production processes and identify waste	Visualize patient treatment process, identify bottlenecks
2	Spaghetti Chart	Illustrate movement of workers and parts to identify inefficiencies in layout	Illustrate staff and patient movement to reduce wasted steps
3	Standardized Work	Standardize steps so all workers follow the same procedures	Standardize operating rooms and treatment flows to reduce errors
4	Root Cause Analysis	Investigate root causes of machine failures or product defects	Identify root causes of medical accidents or errors to prevent recurrence
5	Brainstorming	Generate ideas on production issues and efficiency	Exchange ideas to improve healthcare services and patient care
6	Daily Management	Monitor progress daily and detect issues early	Morning meetings to share information and manage patient care progress
7	Continuous Improvement(Kaizen)	Gather improvement suggestions for production processes	Improve work and service based on feedback from healthcare staff
8	5S	Organize, tidy, clean, standardize, and discipline production lines	Organize work areas in hospitals, manage medication and tools
9	Error Proofing (Poka-Yoke)	Design systems to prevent human errors	Manage medical equipment and medication to prevent mistakes
10	Cell Production	Divide tasks into cells for efficiency	Group patient care into teams or small groups for faster service
11	Just-In-Time (JIT)	Supply only what is needed when it is needed	Ensure medication and tools are available on time
12	Kanban	Visually manage timing of inventory replenishment	Prioritize patient care and manage workloads

workflow and safety. The Spaghetti Chart, shown in Fig. 5, provided insights into delay-prone areas and redundant movements.

- **Root Cause Analysis:** This systematic methodology investigates underlying causes of operational inefficiencies. Widely applied in aviation, medical, and industrial sectors, Root Cause Analysis prioritizes long-term problem resolution over superficial fixes. An example of the Root

Cause Analysis findings is illustrated in Fig. 6, demonstrating a structured approach to diagnosing and addressing core inefficiencies.

By integrating these Lean Thinking methodologies, the study establishes a framework for sustainable operational improvements in MRI workflow management, promoting efficiency, reducing waste, and enhancing overall healthcare service quality.

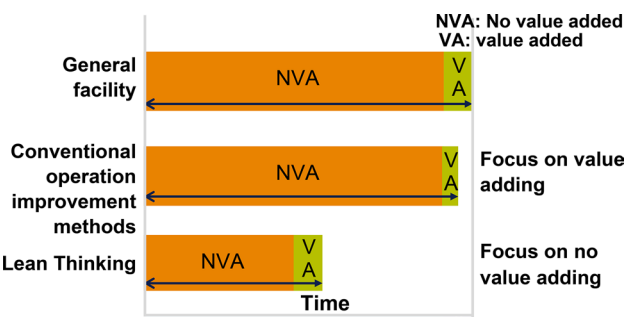


Fig. 1 Lean thinking and conventional operations improvement

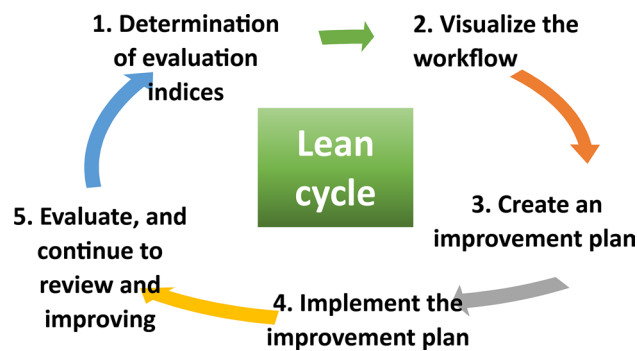


Fig. 2 Lean thinking cycle diagram

Table 2 Lean thinking action plan

	Step 1	Step 2	Step 4	Step 5
	assessment	Current Survey and Improvement Plan	Implementation of improvement plans	Post-improvement plan evaluation and review
action plan	evaluation indices Grasping the problem	Current situation survey and improvement plan. Investigate problems and root causes to gain common understanding. Consider an improvement plan for the obtained problems.	Implement the resulting improvement plan.	Perform post-improvement plan evaluation and review. [Daily Management] Create an evaluation board that serves as an index. Execution of daily management.
output data	Value stream map (Current) Current problem Request	Tools used for the 5 visualizations. Value stream map (Future) Standard work Spaghetti chart Root cause analysis Brain storming	Environmental improvement	Evaluation board that serves as an index Sustainable improvement plan

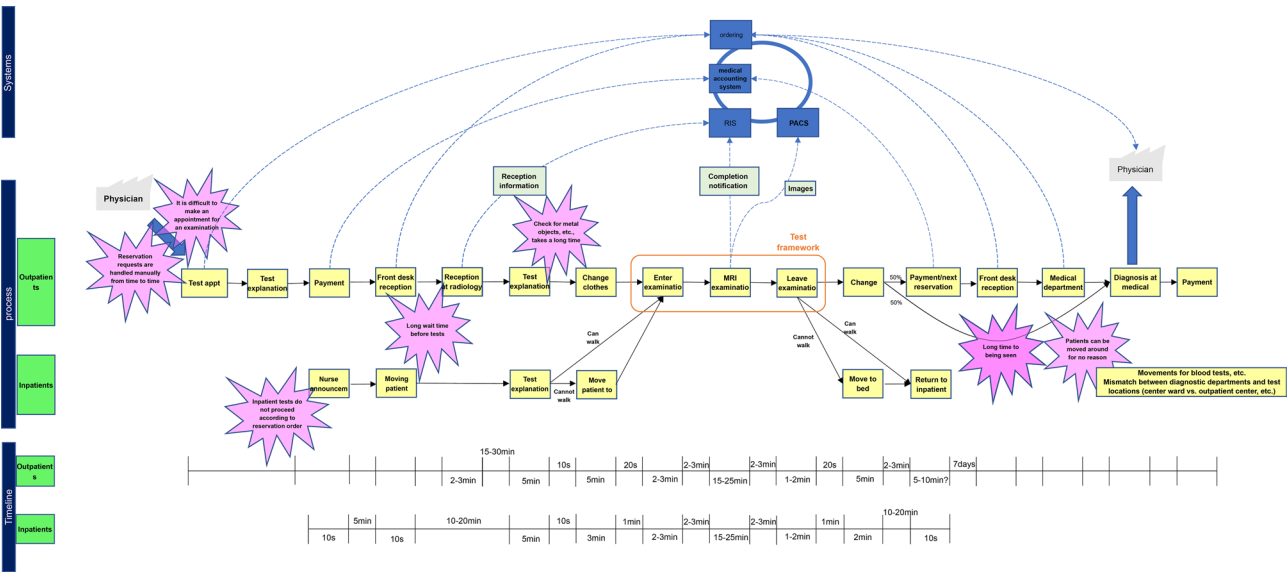


Fig. 3 Value stream map

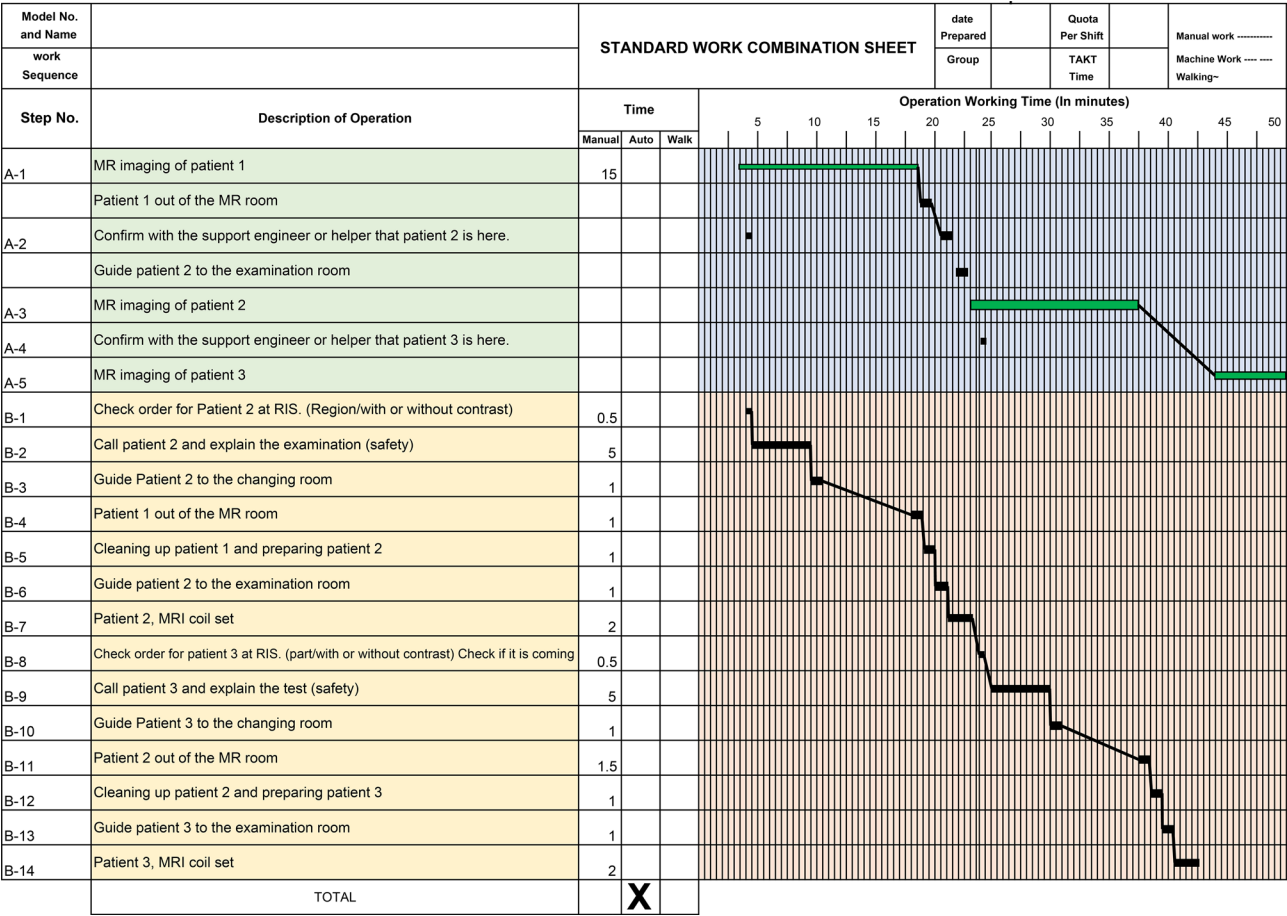


Fig. 4 Standard work

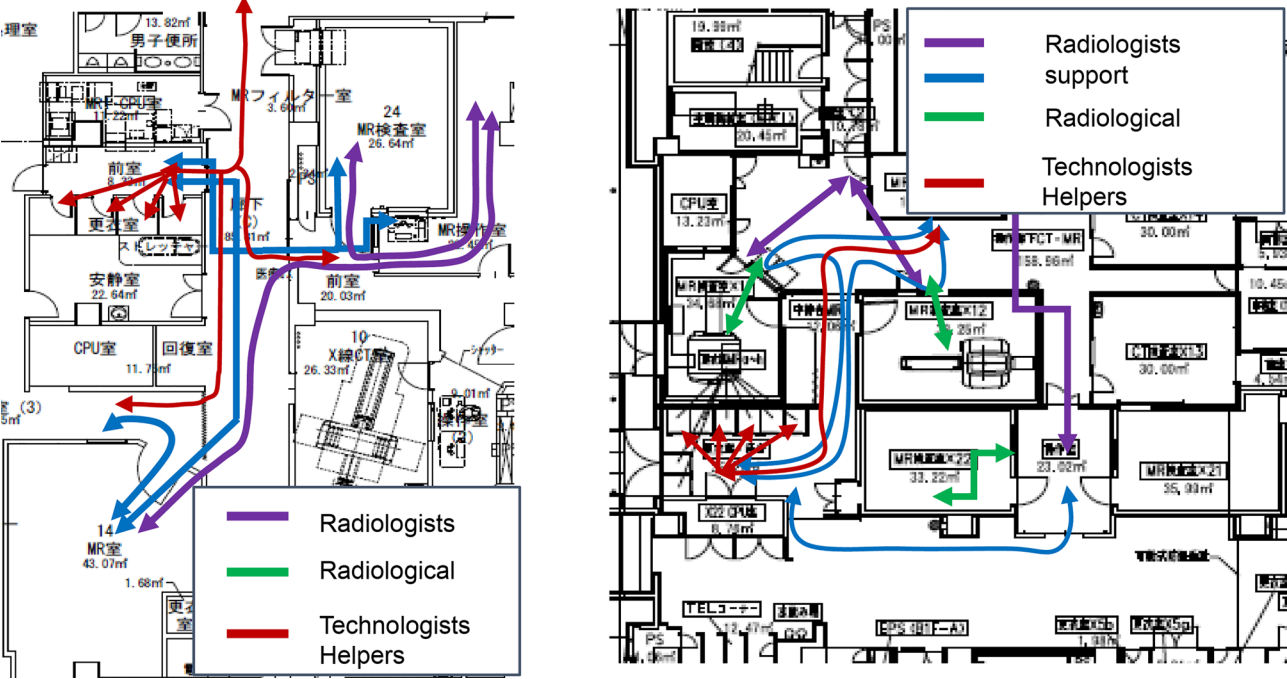


Fig. 5 Spaghetti chart

problem	Patient does not show up for appointment.	Unable to make an appointment for an emergency examination.
root cause	Not properly briefed on going to the hospital 30 minutes prior.	Emergency MRI examinations are available by phone only.
	They do not understand why patients come to the hospital at 30 minutes.	The requesting department calls the laboratory directly to request the examination.
	Affected by weather.	Upon receipt of the request, the laboratory consults with the physician and coordinates with the office.
		One technician will be unavailable to respond to emergency MRI test requests.
solution	Updated the explanation sheet output at the time of reservation	Create a PHS for emergency examination reception physicians and unify the reception.
	Eliminate laboratory vacancies by booking multiple patients into one appointment slot.	Consolidate urgent MRI test requests and assign them to the appropriate laboratory.

Fig. 6 Root cause analysis

Data collection method

This study was conducted in three phases based on a workflow improvement project carried out at Tokyo Women's Medical University Hospital.

Phase 1: collection of baseline data

Baseline examination data conducted over one year from January to December 2016 were collected. The data were extracted from the Radiology Information System (RIS).

Eight items were extracted: examination date, reservation time, examination site, reservation location, reception time, examination start time, examination end time, and examination location. These items were analyzed to assess the operational status.

Between January and April 2016, the movements of radiologic technologists, nurses, physicians, and nurse

assistants were randomly plotted. At the same time, the time required for preparation and examination was measured and plotted for each item.

Phase 2: planning of workflow improvements

From March to August 2017, over a six-month period, the baseline data collected in Phase 1 were used to create Lean tools such as Value Stream Mapping, Spaghetti Charts, and Standardized Work.

Using these tools, issues were identified through Root Cause Analysis.

The extracted issues were reviewed by a team of 14 members, including 12 radiologic technologists with 5 to 20 years of experience and 2 engineers specializing in workflow improvement, to develop specific improvement plans.

Phase 3: implementation and evaluation of improvement plans

First implementation period: January to December 2018

During this period, a 15-minute “Daily Management” meeting was held weekly to continuously discuss improvement ideas.

Second implementation period: January to December 2019

Based on the results of the first phase, further improvement plans were developed and implemented.

Planning and implementation of operational improvement measures

Planning operational improvements

In addressing issues within the MRI examination department, brainstorming sessions were held based on analyses using four key tools: the Value Stream Map, Standard Work, Spaghetti Chart, and Root Cause Analysis. This collaborative effort led to the development of a series of improvement measures aimed at enhancing operational efficiency, which were centered on five main areas:

1. Increase in examination reservations.
2. Rebuilding the team structure.
3. Development of a progress management system.
4. Development of a mirror monitor that allows mutual confirmation of scanning progress.
5. Other operational efficiency improvements.

Improvement plan details

The details of the operational improvement measures planned and implemented in this study are as follows:

- 1) Increasing Examination Reservation Slots by 34%.

By evaluating the current state of examinations and identifying time slots with vacancies, the number of available reservation slots was increased by 34%.

Before the operational improvements in 2016, the weekly reservation slots were 505 (2020/month). After the improvements in 2018, this number rose to 635 (2540/month), a 26% increase. In 2019, the number increased further to 678 slots (2712/month), representing a 34% rise. This growth in slots was achieved in stages.

Specific actions taken to increase the number of slots included:

- I. Adjusting the standard examination duration. Previously, all examinations had a uniform slot duration of 30 min. By optimizing based on actual examination times, the duration for brain/spine/plain exams was reduced to 20 min, and whole-body scans were allocated 30 min, increasing weekly slot availability by 61.

- II. Modifying the scheduling structure. Initially, each slot was assigned to a single patient, but with many no-shows, this was adjusted to allow two patients per slot, maximizing use of available slots. This resulted in an increase of 92 weekly slots.
- III. Shifting examination start times. By advancing the start time by 30 min, the overall daily examination time increased, contributing to an additional 20 slots per week.
- IV. Reducing and releasing fixed reservation slots. Fixed slots for specific departments were cut by two-thirds, and those that remained unused by their reservation deadlines were reallocated to general reservations. Consequently, the average daily number of fixed slots dropped from 15 to 5.

- 2) Productivity improvement by rebuilding the team structure.

Actively utilizing information technology (IT) through brainstorming, as mentioned above, has improved individuals' productivity, even if the number of radiological technicians is reduced and more examinations can be completed during the same working hours.

Therefore, we reduced the number of radiological technicians from 12 to 9 and transferred them to other examination sections to improve the vacation utilization rate and strengthen the education system by assigning a person in charge of educational guidance.

Specific measures for staff reduction are shown below:

- I. We used IT tools such as a centralized progress management system and a mirror monitor, which will be described later. Therefore, we rebuilt the system to include two radiological technologists per MRI unit. As shown in Table 3, the system has been changed to consist of an examination operator in charge of each MRI examination and a controller who understands the entire MRI examination area and provides instructions to the staff.

Based on this measure, productivity per person has improved significantly, and the number of radiological technicians has reduced from 12 to 9.

- 3) Development of a centralized progress management system.

We developed a centralized progress management system to improve the productivity of the radiological technician team in the MRI department.

With conventional radiological information systems, it was unable to check detailed information such as how many patients have already completed the reception

Table 3 Radiological technologist team structure and role changes

• Conventional Team Structure (12 radiological technologists in total) - General Outpatient Center (4 MRI units): 8 radiological technologists (2 radiological technologists/unit) - Central Ward (2 MRI units): 4 radiological technologists (2 radiological technologists/unit)
• Team Structure After Improvement(12 radiological technologists in total) - General Outpatient Center (4 MRI units): 6 radiological technologists (4 operators, 1 controller, 1 libero) - Central Ward (2 MRI units): 3 radiological technologists (2 operators, 1 controller)
Operator: Responsible for MRI examination operations Controller: Understands the entire examination room and issues instructions to staff Libero: Assists the operator

Table 4 Functions of the centralized progress management system**[Functions of the centralized progress management system]**

Overview: Receive information from the radiology information system in real time. •Information such as medical interview, preparation status of MRI examination, and examination in progress can be viewed at a glance.
Upper half of the screen: Displays the progress of MRI examinations currently being performed. •Used for the purpose of sharing information from the operator to the controller. •Displays the progress of the examination, remaining time, physical information, patients scheduled for the next examination. •The remaining time for MRI imaging can be changed in 5-minute increments. The exact remaining time can be shared with the controller. •Displays the next patient name indicated by the controller.
Lower half of the screen: Displays a list of patients currently waiting for examination •Used by controllers to give instructions to operators and other staff. •Displays waiting time, preparation status, next inspection device, etc. As for the waiting time, it is possible to grasp both the elapsed time from the reservation time and the elapsed time from the reception. •The preparation status can be set as interviewing, before needle insertion, preparation completed, or absent, and the controller can grasp the patient's preparation status at a glance.

process, who is undergoing pre-examination interviews, who is available for examination, and how long it will take for patients undergoing examinations to complete. Therefore, it was difficult to understand how work progressed across the entire examination room, and the time from examination preparation to the end of the examination was unnecessarily long. Therefore, we have newly developed a centralized progress management system that receives information from the radiology information system in real-time and confirms in detail the status from reception to the start and end of the examination.

The system extracts the necessary data from database columns in the radiology information system, converts it into information, and displays the information on the screen using a web browser.

In addition to understanding the detailed status of each MRI by utilizing the centralized progress management system, it is possible to understand the patient's preparation status in detail from interviews before injection and completion of examination preparation. Therefore, an

optimal patient allocation for MRI examination based on the situation is possible.

As a result, we could shorten patient waiting times, improve patient satisfaction, and improve the efficiency of the overall operations, thereby creating an optimal examination environment.

Details and screens of the centralized progress management system are presented in Table 4 and Fig. 7.

- 4) Development of a mirror monitor to check the examination status.

Under the new team structure of radiological technologists, as shown in Table 3, we developed a mirror monitor, as shown in Fig. 8, to show the operation screen regardless of which device the controller uses to ensure the same medical safety and quality of examination as before.

The operation monitor screen of each MRI system was distributed and installed next to the monitoring monitor of another MRI system to confirm both operation

[illegible]

Fig. 7 Centralized progress management system screen

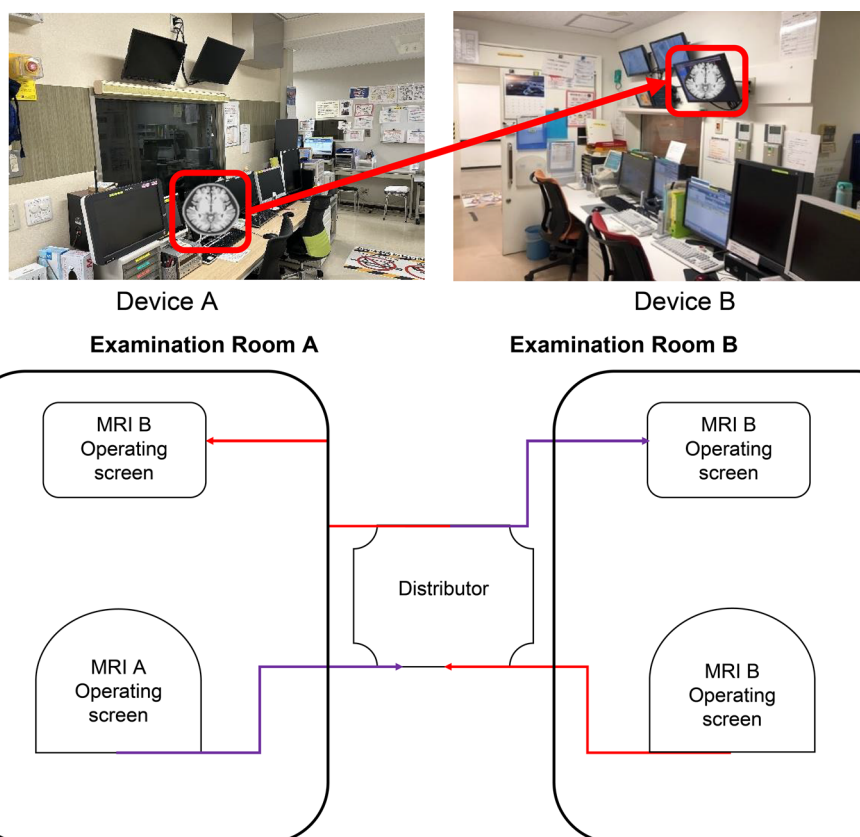


Fig. 8 Mirror monitor configuration diagram

monitor screens. This makes it possible for the controller to check the operation monitor screen from the MRI room and to check the progress, sequence, and operation status remotely.

In addition, depending on the personnel assignment, operators who are unfamiliar with MRI operations or operators in the MRI training process may be assigned. Even operators at a distance can now check for incorrect

operations, missing MRI sequences, or missing clinical findings.

5) Other operational efficiency improvements.

To reduce the workload per person owing to the increase in examination reservation slots and the reduction in staff, various efficiency measures, other than the first to

fourth described so far, have been developed and implemented. The specific measures are as follows:

- I. A manual was created to standardize the pre-examination interview to ensure all examination staff was in charge of the pre-examination interview before entering the room. The training was conducted using a manual that summarizes how to use the pre-examination interview form, the order of interview, important points during the interview, and how to communicate to ensure that not only radiological technicians but also nurses and nursing assistants can conduct pre-examination interviews in the same manner.

In addition, the pre-examination interview is important to confirm that you can enter the MRI room. Considering this opportunity, we examined whether there were any ambiguous expressions or missing necessary items and revised the pre-examination interview form.

- II. The emergency examination coordinator changed from a radiological technologist to an emergency radiologist.

Previously, when an emergency examination was required, the clinical department staff would call the radiology department directly, and the radiological technologist would have to spend time on coordination tasks, thus delaying overall examination operations.

By setting up an emergency radiologist and changing the response to take charge of coordinating emergency examinations, radiological technologists' time is no longer considered, and examinations are not delayed.

- III. We increased the number of daily examinations by approximately five using vacant slots because of reliable cancellation notifications.

Previously, examination cancellation notifications were stopped at the hospital reception desk, and the radiology department could not capture them; therefore, the vacant slots due to cancellations could not be used effectively. By making it a rule that when a cancellation is received at the reception desk of the hospital, the receptionist will contact the examination room, and it is possible to perform examinations ahead of schedule for the waiting patients and to accept emergency examinations, making it possible to increase the number of examinations by approximately five daily by effectively utilizing the cancellation slots.

Daily management

After implementing the improvement plan, the process should not be completed. Instead, daily management should be implemented, which is a key activity to continue evaluation and review, identify further problems and areas for improvement, and continue repeating the lean thinking cycle. In the present study, we held a weekly 15-min meeting (Daily Management) and used a dedicated sheet (Action Tracker), as shown in Fig. 9, to discuss daily management and further improvement plans.

In this study, the project leader prioritized raising awareness and motivating staff through daily management, focusing on cultivating a proactive approach to problem-solving. Team members were encouraged to openly share challenges and issues during meetings, creating an environment where not only the leader but also each staff member could voice their concerns and contribute solutions. The leader also assigned younger staff to take responsibility for specific issues, encouraging them to take an active role in problem-solving as stakeholders. This approach fostered a collaborative culture where staff saw themselves as vital contributors to operational improvements and embraced the MRI department's challenges as their own.

In daily management, in addition to the evaluation indices set at the initial stage, the following indices were added to identify detailed problems, and evaluation and analysis were continuously performed....

1. Total number of examinations per week.
2. Number of examinations per week based on modality.
3. Examination time based on anatomical parts.

Leadership and culture

In this study, leadership was instrumental in creating the conditions under which Lean initiatives could be implemented and sustained in daily MRI operations. Department leaders provided clear direction, ensured access to the necessary resources, and actively participated in the improvement cadence, which fostered a culture of transparency and continuous problem-solving.

Practically, leadership established short, regular Daily Management huddles focused on data (slot utilization, wait days, external referrals) and rapid follow-up. Each action item had an owner and due date, and progress was reviewed the following week, turning meetings from reporting into decision-making forums.

Leaders also delegated ownership to younger staff for specific improvement streams (e.g., pediatric sedation MRI scheduling, breast MRI protocol coordination). This not only accelerated local problem-solving but functioned as on-the-job leadership development, expanding the cadre of people capable of running Lean routines.

ACTION TRACKER					
Date identified	Issue Description	Corrective Action(s)	Person Responsible to Complete	Due date	Status

☐ Action assigned
 ☐ Action complete
 ☒ Root cause fix confirmed

Fig. 9 Action tracker

Cross-functional collaboration was formalized: MRI technologists, nurses, and administrators met around the same visual data (the centralized progress board), which reduced handoff loss and siloed optimization. The mirror monitor arrangement enabled experienced staff to supervise remotely and coach in real time without compromising safety.

While the importance of leadership and culture for Lean sustainability is well recognized in prior research, our findings add a setting-specific insight: aligning leadership engagement with major capital change (e.g., the installation of a new MRI system) created a shared sense of renewal and purpose, which measurably increased staff motivation and accelerated adoption of new work standards [20].

Finally, leaders made failure-tolerant experimentation explicit (e.g., trialing dual-booking where appropriate and revising start times), with pre-defined rollback criteria. This safeguarded patient safety while allowing rapid learning cycles an essential cultural feature for sustaining Lean beyond a one-off project.

Results

Number of examination reservation slots and examinations performed

The improvement plan implemented in this study resulted in an increase in the number of reservation slots.

In 2016, before the operational improvement plan was implemented, the number of reservation slots was 2,020 per month, while after the plan was implemented, it increased to 2,540 per month, indicating a 26% increase (Fig. 10). Continued improvements through daily management led to a further increase in the number of reservations, reaching 2,712 per month in 2019, which represented a 34% increase compared with the pre-improvement period.

Similarly, the number of examinations performed also increased following the operational improvement.

In 2016, the number of examinations performed was 1,850 per month, and after the improvement plan was implemented in 2018, it rose to 2,210 per month, reflecting a 19% increase.

By 2019, the number of examinations reached 2,360 per month, representing a 28% increase compared to the pre-improvement period. (Fig. 11).

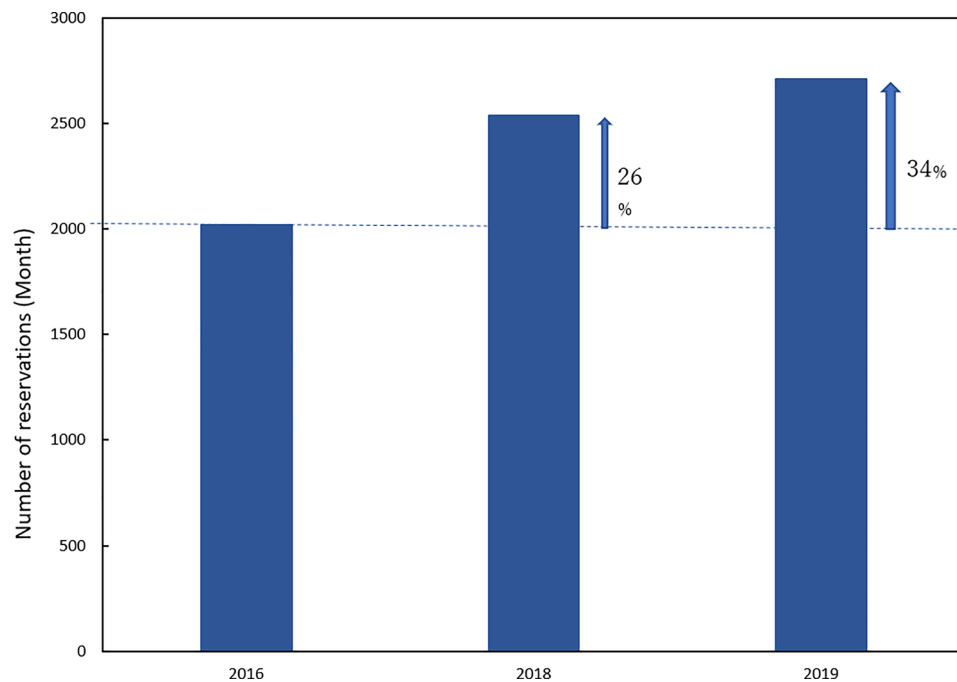


Fig. 10 The number of reservation slots per month

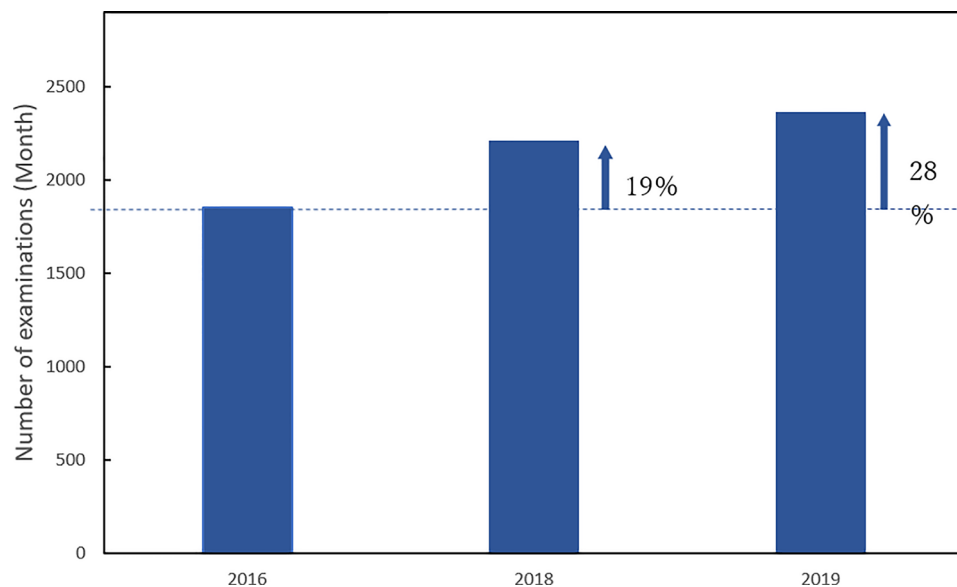


Fig. 11 The number of examinations performed monthly

Waiting days for reservation

The implementation of the improvement plan resulted in a significant reduction in waiting days for reservations. In 2018, the average waiting days for reservations decreased from 45 days to 10 days, representing a 78% reduction from the original value, as shown in Fig. 12. By 2019, continuous improvements led to a further reduction in waiting days to an average of 5 days, representing an 89% reduction from the initial waiting period.

For specific types of exams such as brain MRI, brain MRI+MR-angiography (MRA), contrast MRI, gallbladder pancreatic Magnetic Resonance Cholangiopancreatography (MRCP), and whole-body MRI, which collectively account for more than 70% of the reservation slots, it became possible to reserve and perform examinations as early as the next business day. This significantly contributed to the reduction in waiting days.

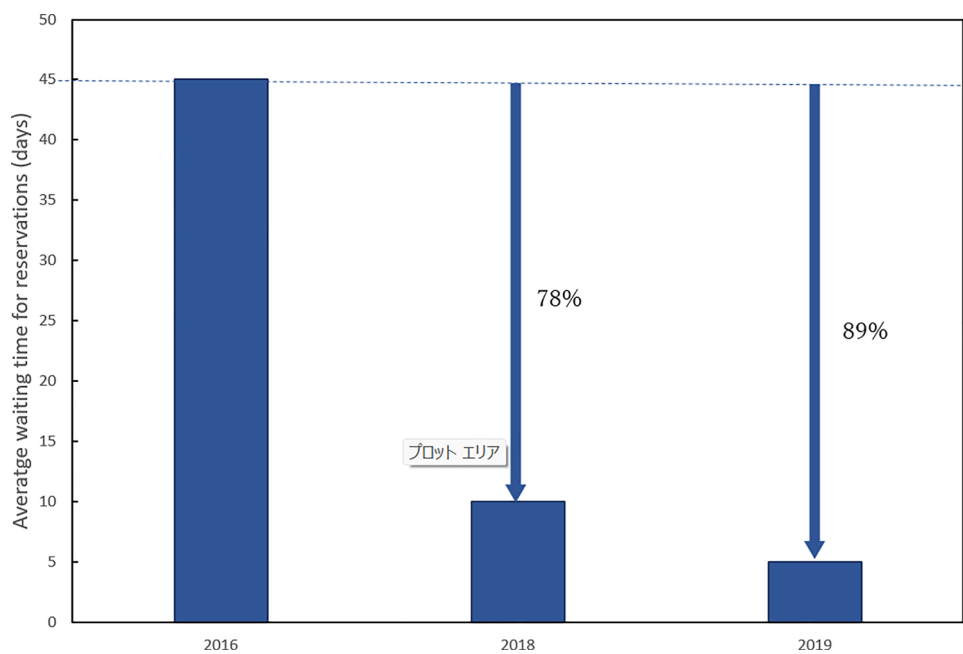


Fig. 12 The number of waiting days for reservation

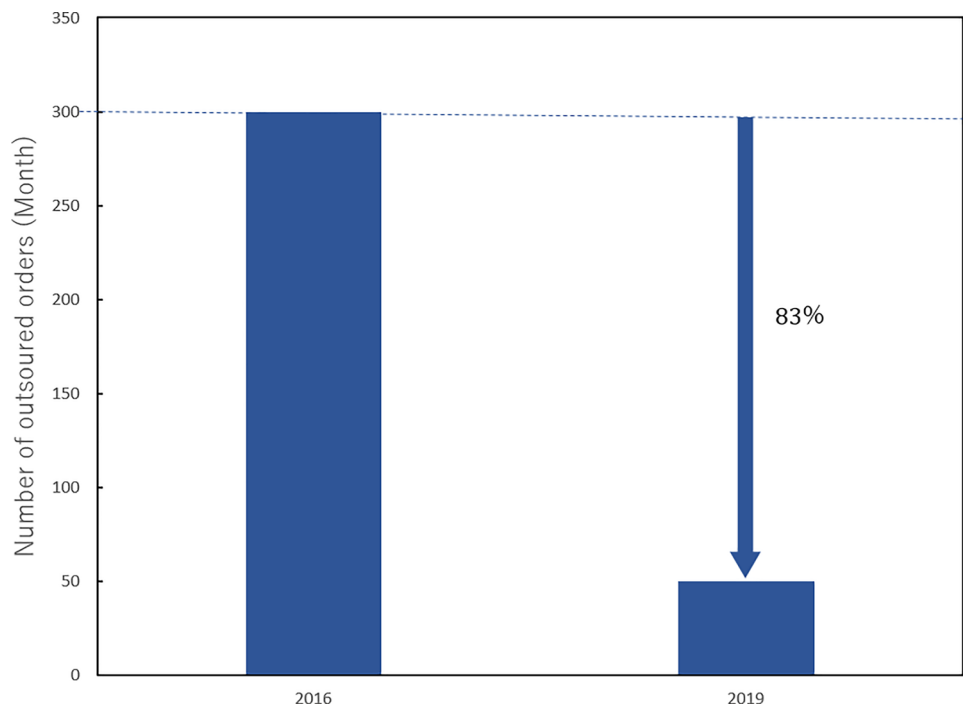


Fig. 13 The number of examination orders to other facilities

Number of examination orders to other facilities

The reduction in waiting days for reservations and the increase in the number of examination reservation slots and performed examinations contributed to a significant decrease in the number of outsourced tests.

As shown in Fig. 13, the number of outsourced tests decreased from an average of 300 per month to 50 per

month, representing an 83% decrease from the previous level.

Discussion

Theoretical foundation and conceptual integration

To frame the intervention more broadly, the results of this study can be understood through the lens of an

integrated Business Process Reengineering (BPR), Lean Thinking, and Information and Communication Technology (ICT) framework. As shown in Figure xx, BPR focuses on radical redesign of core processes, while Lean emphasizes continuous waste elimination and staff-driven process improvement. When these two methodologies are combined with ICT which enables real-time data visualization and operational coordination a powerful synergy is created. This integrative model aligns with recent healthcare transformation paradigms that emphasize both structural redesign and iterative improvement supported by digital solutions. In our study, this approach allowed for system-wide optimization rather than isolated enhancements, supporting both efficiency and adaptability Fig. 14.

Increase in the number of examination reservation slots and examinations performed

In this study, the implementation of an operational improvement plan using Lean Thinking led to a 26% increase in the number of examination reservation slots and a 19% increase in the number of examinations performed in 2018 compared to before the operational improvements. Further improvements made in 2019 led to an increase of 34% in reservation slots and 28% in examinations performed compared to 2018. A key characteristic of Lean Thinking is that operational improvements are not static; rather, they are sustained through the continuous repetition of the Lean cycle. In 2019, this approach resulted in an additional increase of eight and nine points, respectively, in reservation slots and examinations performed.

Continuous operational improvement is supported by daily management, which is discussed further below.

Reduction in the number of waiting days for reservations and outsourcing

Operational improvements significantly reduced the number of waiting days for reservations. In 2018, the

average waiting time for reservations decreased to 10 days. This reduction was primarily attributed to the ability to reserve and perform examinations as early as the next business day, including procedures such as brain MRI, brain MRI + MRA, contrast MRI, gallbladder pancreatic MRCP, and whole-body MRI. Further analysis through daily management highlighted that waiting times increased due to limited reservation slots for breast MRI and pediatric sedation MRI. By increasing the number of available slots for breast MRI, reassessing pediatric sedation MRI slots, and reallocating them as necessary, it was possible to reduce waiting times to an average of five days by 2019.

Despite these improvements, some challenges remained. For instance, when no examination slots were available on a particular day, the next available reservation would be the following week. Additionally, overlapping holidays and vacations still led to extended waiting times.

Regarding outsourced examinations, the number of outsourced tests decreased significantly as a result of the increased number of reservations and examinations performed in the hospital.

The number of outsourced examinations dropped from 300 per month to 50 per month, representing an 83% decrease.

Most of the outsourced examinations were performed at affiliated hospitals, and therefore, outsourcing to unaffiliated hospitals was effectively eliminated. The reduction in waiting days and outsourced examinations contributed to improved satisfaction levels across clinical departments.

Productivity in MRI departments

The implementation of Lean Thinking in the MRI department resulted in a 28% increase in the number of examinations performed compared to pre-improvement levels, leading to a corresponding increase in productivity. The number of radiological technologists was reduced from 12 to 9, thanks to a restructured team. This reduction in staff contributed to a 33% improvement in productivity. Taking the reduction in radiological technologists into account, overall productivity increased by 71%.

While staff reductions typically result in decreased productivity, the improvements implemented in this study not only mitigated this effect but also contributed to a significant rise in the number of examinations performed. A key factor was the centralized progress management system, which allowed for monitoring patient preparation and MRI room progress—tasks that were previously difficult to track using the radiological information system. This system also enabled optimal patient assignment based on real-time circumstances.

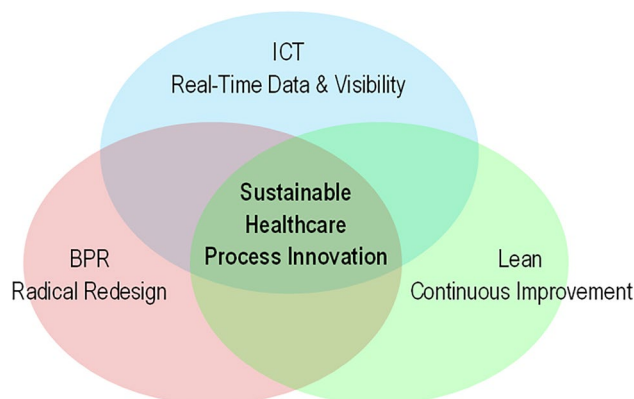


Fig. 14 Conceptual framework model of BPR, lean thinking, and ICT

Additionally, the installation of mirror monitors enabled skilled staff in both MRI rooms to oversee examination progress and assist less experienced operators. This created a support system that ensured safe examination procedures, improved productivity, and contributed significantly to medical safety.

Traditionally, Lean initiatives aimed at improving productivity in radiology departments have been undertaken exclusively by internal medical staff. However, such efforts are often limited to adjustments in staff allocation and task reorganization, restricting the scope of potential operational improvements [2, 21, 22].

In contrast, this study adopted a collaborative approach by involving external MRI system and ICT vendors from the early stages of the Lean process. Their technical expertise enabled the development and implementation of innovative solutions such as a centralized progress management system and real-time mirror monitoring that extended beyond the scope of conventional staff-led improvements. This multidisciplinary collaboration significantly broadened the impact of Lean implementation and contributed to substantial gains in operational efficiency.

However, Lean implementation in healthcare settings is often accompanied by several inherent challenges, including staff resistance, cultural incompatibility, sustainability of improvements, maintaining patient centered care, and resource limitations.

In our study, we proactively addressed these challenges. To mitigate staff resistance, we actively engaged radiological technologists and support staff from the early stages of Lean implementation. Staff were invited to participate in brainstorming sessions, and their input was incorporated into the improvement plans. This approach fostered a sense of ownership and reduced skepticism, making the Lean process more collaborative.

To overcome organizational cultural barriers, we secured strong top-down support from hospital leadership and formed cross-departmental teams to dissolve siloed communication. This promoted a culture of transparency, continuous improvement, and collective problem-solving.

To ensure sustainability, we standardized new workflows and incorporated continuous monitoring through Daily Management. Key performance indicators (KPIs) such as number of examinations and waiting time were tracked weekly using the Action Tracker. This allowed for ongoing evaluation and timely corrective actions. In addition, younger staff were assigned responsibility for specific issues, cultivating future leaders in Lean operations.

Patient-centeredness was maintained by including patient satisfaction and examination accessibility as core outcome metrics. By shortening waiting times and

improving appointment flexibility, we enhanced both operational efficiency and the quality of patient experience. Feedback from patients was also considered during the redesign of pre-examination processes.

Lastly, despite limited resources, we utilized ICT solutions such as the centralized progress management system and mirror monitors to improve productivity without increasing manpower. This technology-enabled approach was key to sustaining improvements without placing additional burden on staff.

Additionally, this study uniquely highlights that aligning Lean initiatives with the timing of major capital investments such as the introduction of a new MRI system can significantly enhance team motivation and shared commitment. This strategic alignment created a shared vision among stakeholders, increasing engagement and ownership of Lean activities. By implementing Lean improvements concurrently with the installation of a new MRI, staff members perceived a sense of renewal and progress, fostering an environment conducive to operational innovation.

We believe these practical strategies for overcoming Lean-related challenges can be generalized and applied to other medical departments aiming to adopt Lean methodologies. Sharing such lessons may aid in broader implementation and adaptation across healthcare institutions.

Comparison with global Lean/BPR healthcare projects

The operational improvements achieved in this study such as reducing MRI reservation wait times from 45 to 5 days and cutting external referrals from 300 to 50 per month are highly significant by global standards. For example, Virginia Mason Medical Center in the United States applied Lean hospital-wide and reported an 85% reduction in diagnostic turnaround time and over \$2 million in annual inventory savings [23]. In Denmark, Lean initiatives in radiology cut CT waitlists from 12 to 4 weeks and boosted daily throughput by six additional exams without added staffing. Table 5 presents selected international cases of Lean/BPR-based interventions in healthcare, comparing their strategies and outcomes [20].

Daily management

In this study, weekly meetings known as Daily Management were conducted for approximately 15 min to systematically apply the Lean Thinking cycle. During these meetings, the current status following the implementation of operational improvements was assessed, and further issues as well as potential areas for enhancement were identified. This structured approach enabled a continuous cycle of improvement and evaluation.

To gain a comprehensive understanding of the operational status of the MRI department, several indices were

Table 5 Comparison with global Lean/BPR healthcare projects

Country	Medical Area	Methodology	Outcomes
USA	Multidepartment	Lean (TPS)	↓85% turn-around time, ↓\$2 M inventory costs
Denmark	Radiology (CT)	Lean (5 S, Kaizen)	↓ Waitlist 12→4 weeks, ↑ throughput
Spain	Surgery	Lean BPR	↓ LOS 43%, ↑ surgical capacity
India	Outpatient clinics	Lean (VSM)	↓ Wait time 60→15 min
Saudi Arabia	Operating Rooms	Lean process redesign	↑ Start time punctuality, ↓ turnover delays

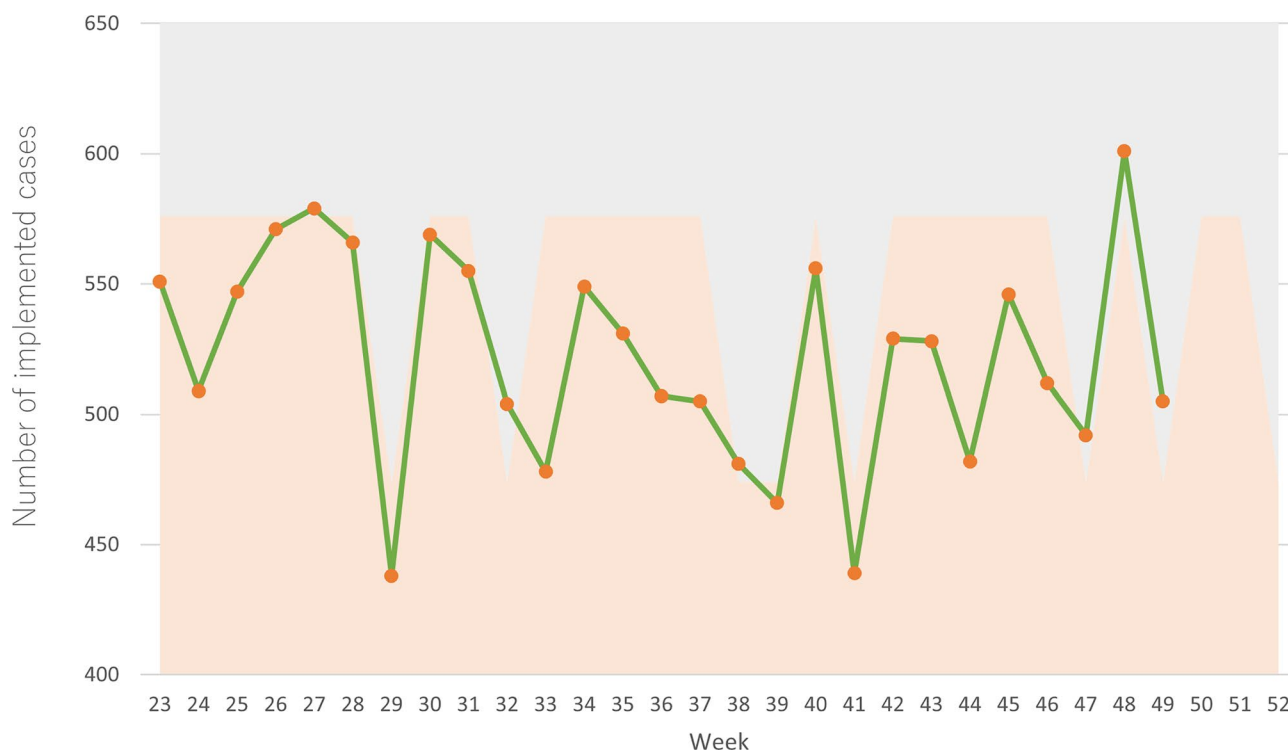
examined, including the total number of examinations performed per week, the total number of examinations conducted in each MRI room per week, and the examination time categorized by anatomical parts. Graphs representing these indices were constructed to facilitate problem identification and analysis.

For example, Fig. 15 illustrates the total number of examinations performed within one week compared to the number of scheduled reservation slots. A noticeable decrease in the number of examinations relative to the total number of reservations was compared with other

indices to determine whether the cause was sudden or required a long-term solution. This approach allowed for prompt identification of issues and formulation of appropriate improvements.

Figure 16 presents the number of examinations conducted per week across various MRI rooms over 42 weeks in 2018. This data enables an understanding of the operational status of each MRI room, which is further analyzed through comparison with factors such as equipment failures, protocol changes, operator skill levels, and extensions due to special procedures during examinations. Such composite analysis facilitates the identification of root causes. For instance, during the 42nd week of 2018, the number of examinations performed in the C, D, and F MRI rooms was significantly lower than planned, suggesting modality-specific issues that required further investigation.

The examination time for each anatomical part is illustrated in Fig. 17. An increase in brain MRI examination time was observed during the 43rd week. Analysis revealed that the cause of this increase was the introduction of a new protocol requested by the clinical department, which led to extended examination times due to inexperienced operators and incorrect protocol settings. By revising the protocol, establishing appropriate protocols for all MRI procedures, and ensuring operators were properly informed, examination times improved

**Fig. 15** Total number of examinations per week (2018)

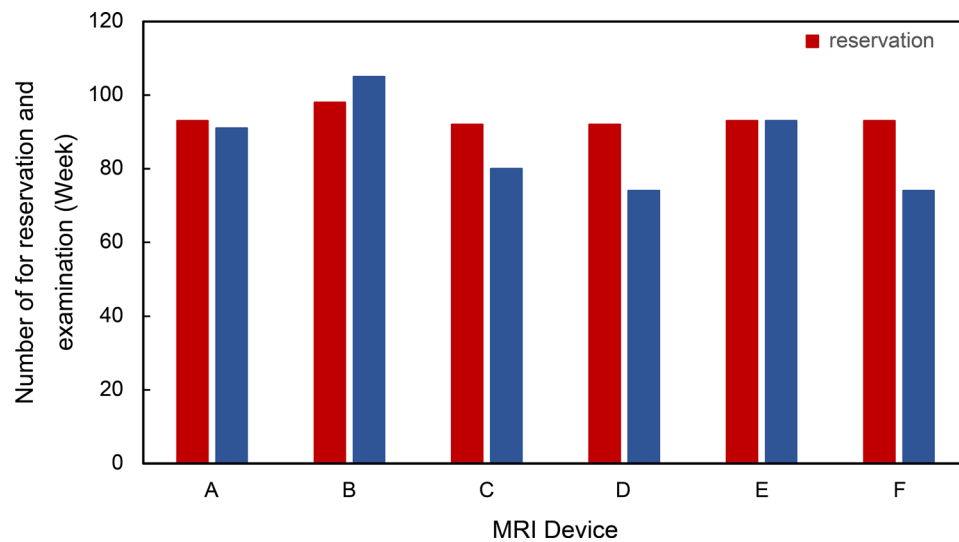


Fig. 16 Number of weekly examinations performed by MRI room (42 weeks in 2018)

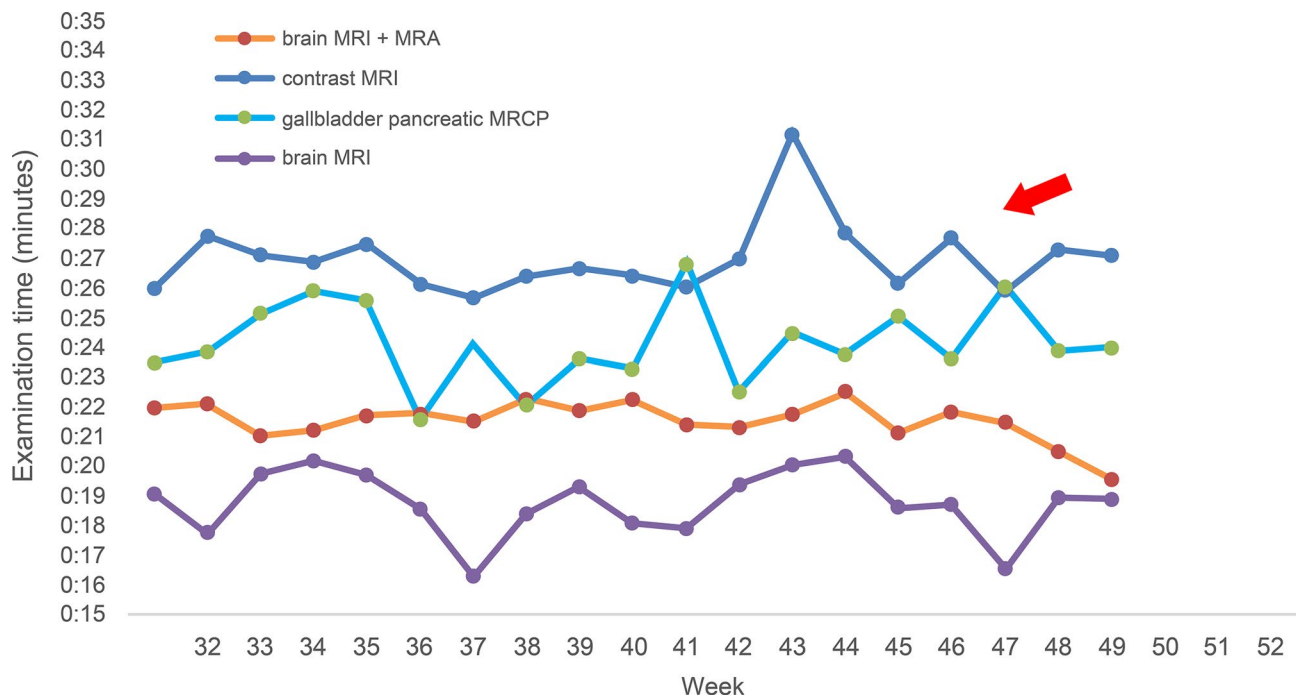


Fig. 17 Examination time according to site (2018)

significantly the following week, allowing examinations to be conducted within the normal time frame.

Lean tools such as Value Stream Mapping, Spaghetti Diagrams, Standardized Work, and Root Cause Analysis all of which were utilized in this study are widely recognized in healthcare process improvement for their ability to visualize processes and identify inefficiencies. Each tool offers distinct advantages but also comes with certain limitations as reported in the literature. For example, Value Stream Mapping provides a comprehensive visualization of the entire patient workflow and helps uncover

wasteful steps, yet it can be time consuming and its output may depend on the accuracy of collected data and the perspective of the analysts involved. In healthcare settings, one limitation of VSM is that it may be difficult to apply consistently due to variations in patient conditions and needs. Similarly, Spaghetti Diagrams effectively map staff and patient movements to reveal unnecessary motion in the workplace, thereby highlighting opportunities to streamline physical flow. However, a spaghetti map alone does not convey the time or frequency of each movement, and complex care environments can

make such diagrams cumbersome to interpret without additional context. Standardized Work is a Lean method that ensures each task is performed in a consistent, best-practice manner, which can improve safety, quality, and efficiency in healthcare. The challenge with standardized workflows is maintaining flexibility: strict standardization can be counterproductive if it doesn't allow adaptation to individual patient needs, so it should be created by those who do the work and serve as a basis for continuous improvement rather than as an inflexible rule. Root Cause Analysis (RCA), including techniques like the "5 Whys," provides a systematic approach to dig deeper into underlying causes of operational problems. This tool's strength is in preventing superficial fixes by addressing fundamental issues; however, its effectiveness relies on thorough investigation and an unbiased, multidisciplinary perspective. If an RCA is rushed or performed with narrow viewpoint, it can lead to misinterpretations or incomplete solutions. Notably, no single Lean tool is sufficient by itself the tools are most powerful when used in combination to provide complementary insights. In our project, this combination of VSM for mapping the overall process, spaghetti diagrams for spatial flow, standardized work for task consistency, and RCA for problem-solving ensured a holistic understanding of the MRI workflow.

In this study, we also took deliberate measures to prevent errors, misinterpretation, or inappropriate actions when applying these Lean tools. First, analyses and mapping exercises were conducted by a multidisciplinary team of stakeholders (including radiological technologists, nurses, and administrators) rather than by individuals in isolation. Involving an interprofessional team in the analysis provided multiple viewpoints and helped minimize personal bias, which aligns with recommended practices for thorough root cause analysis. This collective review ensured that the Value Stream Map and Spaghetti Diagram accurately reflected the reality of the MRI unit's operations, and that conclusions drawn from them were agreed upon by various professionals. Second, we established standardized procedures for how each tool would be used in the project. For instance, the team followed a step-by-step protocol for creating the Value Stream Map (defining process steps, collecting data, validating the map) and for conducting RCA (using a structured "5 Whys" approach with verification of each causal assumption). By standardizing the way we applied these tools, we reduced variability in the analysis process itself and made the improvement workflow transparent and repeatable. Third, we provided training and education sessions on Lean methodology and the specific tools to all team members involved in the project. This ensured that everyone had a clear understanding of how to use and interpret the tools correctly for example, training staff on how

to track their activities for the spaghetti diagram and how to identify waste in the flow, as well as educating them on the goals of standardized work documentation. Finally, any findings and proposed improvements identified with the tools were subject to multi disciplinary review and validation during our Daily Management meetings before implementation. In these brief weekly meetings, the team jointly examined the proposed changes (such as adjustments in scheduling or workflow) to verify that they were supported by data and frontline experience. This step created an additional safeguard against misinterpretation, as any questionable analysis result would be questioned and clarified by the group. Through these combined precautions team based analysis, standardized tool usage, staff training, and collective review we aimed to prevent missteps and ensure that Lean interventions led to genuinely effective and appropriate improvements.

In our study, leadership commitment was reflected in cross-departmental collaboration, empowerment of younger staff, and integration of ICT solutions. These actions linked directly to our results sustained operational improvements, shorter waiting times, and reduced outsourcing demonstrating that engaged, context-specific leadership behaviors drive measurable Lean outcomes.

Furthermore, recent studies have reinforced the central role of leadership and organizational culture in sustaining Lean transformations. For example, strong leadership commitment including clear communication and active involvement has been shown to foster the supportive environment essential for successful Lean initiatives [24]. In healthcare settings, leadership together with an improvement oriented culture can directly drive process improvements and also empower staff to sustain continuous improvement efforts [25]. Conversely, a lack of leadership support or adequate Lean expertise is frequently identified as a barrier to maintaining long-term Lean gains in organizations [26]. These recent findings align with our approach of nurturing a participative, continuous-improvement culture under committed leadership to ensure that the operational gains achieved through Lean Thinking are maintained over time.

To support replication in other healthcare settings, we propose a structured Lean-ICT implementation roadmap consisting of five key phases: (1) strategic alignment with institutional investments, (2) cross-functional team building with external vendors, (3) visualization and diagnosis using Lean tools, (4) implementation of ICT-supported operational solutions, and (5) daily management and KPI tracking for sustainability. This phased roadmap provides a practical and adaptable guide for other institutions aiming to integrate Lean Thinking with digital health infrastructure.

However, when other institutions attempt to implement these improvements, they should be mindful of

potential pitfalls. For instance, if frontline staff do not fully embrace the new processes, improvements may not advance in practice. In addition, the initial cost of introducing the required ICT infrastructure can pose a major hurdle, especially for facilities with limited budgets. Finally, even if such a system is formally introduced, there remains a risk that it becomes a mere formality and is not actively sustained in daily operations.

Generalizability and future research directions

While this study was focused on MRI operations, the integrated BPR–Lean–ICT model has strong potential for generalization to other diagnostic imaging areas (e.g., CT, X-ray, ultrasound) and clinical departments such as surgery, emergency, or outpatient care. International case studies have demonstrated similar improvements using Lean or BPR in various contexts, suggesting that our centralized ICT-supported framework could provide value across diverse hospital settings.

Future research should explore the long-term sustainability of such improvements, especially in maintaining staff engagement and adapting to evolving clinical environments. Moreover, challenges such as user interface complexity of ICT systems, interdepartmental coordination, and initial implementation costs must be addressed. Emerging technologies such as AI-based predictive scheduling and IoT-enabled patient flow tracking—may offer the next stage in process optimization. Investigating the role of digital transformation in Lean healthcare (“Healthcare 5.0”) could provide a fruitful direction for subsequent studies.

Framing results using BPR evaluation metrics

To provide a comprehensive assessment of the intervention outcomes, we further structured our results using the four evaluation metrics commonly employed in Business Process Reengineering (BPR): cost, time, quality, and flexibility.

Sustainability & Cultural Transformation

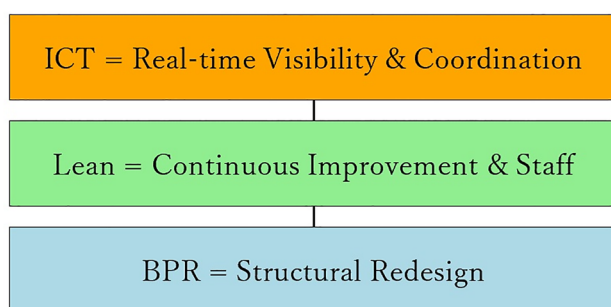


Fig. 18 Conceptual model of BPR–Lean–ICT integration in healthcare operations

Cost: The number of radiological technologists was reduced from 12 to 9 without compromising service quality, resulting in improved productivity and personnel cost optimization. Moreover, the reduction of outsourced MRI examinations from 300 to 50 per month significantly reduced institutional expenditure related to external imaging services.

Time: A key achievement was the reduction of MRI reservation waiting time from 45 days to 5 days. Streamlined workflows and optimized scheduling directly shortened the cycle time from patient referral to examination completion. The ICT-supported centralized progress management system further reduced examination preparation time and improved overall scheduling efficiency.

Quality: Medical safety and examination quality were maintained or improved despite staff reductions, owing to enhanced training, standardization of pre-examination interviews, and real-time monitoring tools such as the mirror monitor system. Patient satisfaction was also improved through shorter wait times and more responsive examination availability.

Flexibility: The new team structure allowed for flexible role allocation, enabling technologists to cover multiple responsibilities depending on real-time needs. The scheduling system was modified to accommodate dynamic slot allocation, including the use of cancellation slots and adjustment for special cases such as pediatric sedation or breast MRI.

This BPR-based evaluation framework reinforces that our intervention not only achieved operational efficiency but also aligned with broader institutional goals for high-value, patient-centered healthcare transformation.

Beyond the immediate operational results, this study advances the theoretical and managerial understanding of Lean healthcare. Conceptually, the BPR–Lean–ICT integration can be viewed as a layered model: BPR provides the structural redesign, Lean ensures iterative staff-driven improvement, and ICT offers real-time visibility and coordination. This tripartite framework, as illustrated in Fig. 18, aligns with established theories of organizational change, such as Kotter’s stages of transformation and Schein’s model of cultural embedding, thereby situating the study within a broader academic discourse.

From a practical standpoint, the key factors identified here – (1) inclusion of IT and vendor experts in cross-functional teams, (2) synchronization of Lean with major capital investment, (3) daily KPI-based management, and (4) visible leadership commitment – as summarized in Table 6, can serve as a reproducible checklist for other institutions. Future projects could apply this checklist to diverse settings such as CT, surgery, or outpatient clinics, adapting the same principles to different operational challenges.

Table 6 Key factors for replicating Lean–ICT interventions in healthcare settings

Factor	Description	Managerial/Practical Value
Cross-functional team including IT/vendor experts	Inclusion of engineers and IT specialists in addition to technologists, nurses, and administrators	Enables innovative ICT-based solutions beyond clinical staff knowledge
Synchronization with capital investment	Align Lean activities with installation of new high-value equipment (e.g., MRI systems)	Enhances staff motivation, creates shared purpose with leadership
Daily management with KPI monitoring	Weekly/short-cycle meetings with visual dashboards and action trackers	Ensures continuous improvement and sustainability
Visible leadership engagement	Leaders participate in daily routines and delegate ownership to younger staff	Builds a culture of accountability and long-term organizational learning

In comparing our work with prior research, some relevant parallels and key distinctions emerge. Akdag, Çalışkan, and Toma (2017) conducted a Lean process redesign in a radiology department, using value-stream mapping to identify and reduce non-value-added processes [27]. While this work focused on workflow diagnostics and Lean tools, it did not report on the inclusion of external technical specialists such as IT or device vendor experts to co-design ICT or equipment-based solutions. Similarly, recent studies on technology integration in nursing and administrative healthcare settings have shown benefits of digitalization for efficiency and documentation (e.g., Technology integration in nursing and healthcare administration, 2019), yet the involvement of specialized device/IT engineers in the improvement team remains rare [28].

The 2025 study by Capacho-Alfonso et al. develops a model for big data and health technology adoption, integrating technical, organizational, social, ethical, and legal dimensions. However, their model is largely at the organizational or policy level rather than grounded in hands-on operational redesign of imaging operations with on-site technical intervention [29].

Our study adds new knowledge in several respects: (1) It empirically demonstrates that including device and IT specialists in the cross-functional Lean team enables solutions (hardware layout changes, ICT dashboards, mirror monitoring, etc.) that clinical staff alone would not conceive. (2) It situates the BPR–Lean–ICT integration as a layered model (Fig. 18), showing how external technical expertise forms a crucial component in achieving sustained productivity and cultural change. (3) It provides a replicable checklist (Table 6) that other imaging departments or diagnostic units can use to guide team composition, investment timing, and leadership

involvement—filling a gap in prior work which often lacks such operational detail.

Conclusions

In this study, we examined the possibility of improving MRI room operations through behavior analysis and Lean Thinking, based on the Toyota Production System. Key to our success was the role of leadership and a supportive culture that empowered team members to engage actively in continuous improvement efforts. Notably, after implementing the improvement plan, the number of examination reservation slots and examinations performed increased by 26% and 19%, respectively, demonstrating a significant enhancement in throughput and patient accessibility. Furthermore, by making detailed improvements through Daily Management, the reservation slots and examinations performed increased by 34% and 28%, respectively, compared to before improvements.

The leadership’s commitment to fostering a collaborative and problem-solving culture in the MRI department was instrumental in sustaining these improvements. By encouraging open communication, giving each team member a voice, and assigning responsibilities to younger staff, the department established a culture where staff actively contributed to identifying and addressing challenges. This cultural shift contributed to a sustainable model for integrating Lean and ICT in healthcare, providing substantial productivity gains while reducing staffing requirements, marking an innovative advancement in lean healthcare research. This study contributes to the healthcare field by demonstrating that Lean Thinking, typically applied in manufacturing, can be adapted to medical settings to enhance operational efficiency.

Specifically, our approach significantly increased the utilization rate of MRI reservation slots, reduced the average waiting days to five, and decreased the need for outsourcing examinations to other facilities, thereby improving service accessibility and clinical department satisfaction.

However, implementing Lean Thinking in a medical context presented certain challenges, such as the need for continuous, detailed adjustments through Daily Management and the complexity of coordinating staff schedules without disrupting patient care. These difficulties required ongoing engagement from staff and a commitment to iterative improvements, highlighting that such operational adjustments may require time to achieve full optimization.

This study highlights how integrating Lean Thinking with ICT contributes to improved hospital management by enhancing efficiency, reducing costs, and optimizing resource allocation. These findings support data-driven decision-making and promote sustainable operational improvements in healthcare institutions.

Our findings demonstrate that Lean Thinking can be effectively applied beyond manufacturing to non-urgent medical settings. Moreover, the integration of Lean Thinking and ICT not only improves operational efficiency but also enhances staff engagement and teamwork, providing new insights for Lean Healthcare research.

The study also highlights two practical considerations that may guide other institutions seeking similar improvements. First, forming cross-functional teams that deliberately include IT and vendor specialists expands the scope of possible solutions and accelerates the adoption of ICT-enabled workflows. Second, synchronizing Lean implementation with major equipment investments provides a powerful mechanism for aligning leadership priorities with frontline practice, thereby enhancing staff motivation and sustaining cultural change.

To facilitate replication, we propose a practical checklist of four key factors (Table 6) and a conceptual model of BPR–Lean–ICT integration (Fig. 18). Together, these managerial insights and theoretical contributions provide a transferable framework that can be adapted to diverse healthcare settings beyond MRI operations.

This study confirmed that Lean Thinking and ICT integration effectively improve MRI operations. However, optimizing the implementation process and assessing applicability to other medical departments remain future challenges. Further research should evaluate long-term impacts, including patient satisfaction and overall healthcare quality improvements.

In addition, future studies should evaluate this framework through multi-center implementations and long-term assessments. Such investigations would provide stronger evidence of generalizability across different healthcare contexts and further clarify its impact on patient-centered outcomes, organizational culture, and healthcare quality at scale.

This study has several limitations. First, the investigation was conducted in a single MRI department at one institution, which may limit the generalizability of the results to other settings. Second, although short-term improvements following the application of Lean methods were observed, further research is needed to assess the long-term sustainability and the impact on staff motivation. Additionally, the effectiveness of the ICT system may vary depending on its operational status and changes in the healthcare environment, warranting more detailed evaluation in future studies."

Future work could explore similar integrations across other departments, assessing Lean's adaptability and operational impact in diverse healthcare settings. Additionally, longitudinal studies are recommended to evaluate the long-term impact of these operational changes on patient satisfaction and healthcare delivery efficiency.

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Author contributions

All authors read and approved the final manuscript.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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